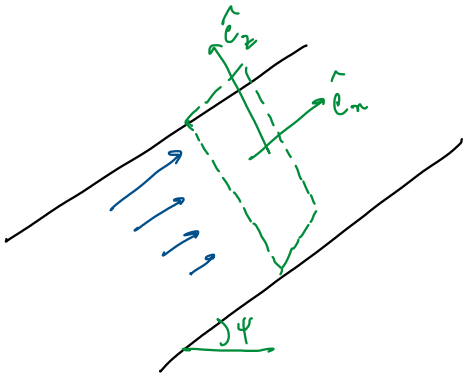
6.39

Sunday, 29 October 2023 6:50 PM



$$\frac{\partial P}{\partial z} = -\rho g_z$$

$$P = -\rho g_z z + f(r) \quad ; \quad \frac{dP}{dr} = \frac{\Delta P}{L} = \frac{df}{dr}$$

$$f(r) = \frac{\Delta P}{L} r + C_1$$

$$P = -\rho g_z z + \frac{\Delta P}{L} r + C_1$$

BC → $P(z, r) = P(0, 0) = P_0 \} \Rightarrow C_1 = P_0$

$$\therefore \boxed{P = -\rho g_z z + \frac{\Delta P}{L} + P_0}$$

$$\frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) = 0$$

Incomp ✓

$$\Rightarrow \rho \frac{\partial}{\partial r}(r v_r) = 0$$

$\frac{\partial}{\partial r}(r v_r) \neq \frac{\partial v_r}{\partial r}$

↳ $\frac{\partial}{\partial r}(r v_r) \neq \frac{\partial v_r}{\partial r}$

Integrate once

$\Rightarrow \quad r v_r = c_1$

$$\boxed{v_r = \frac{c_1}{r}}$$

$$Q = \langle v \rangle \cdot A$$